

Equivalent Expressions: Distribute, Combine, Factor

distributing, combining like terms, and factoring out a common factor

Directions:

- Show every step. An answer is an *expression*, so keep the variable in it.
- In the story problems, the units and givens are stated in the problem — use them.

1. Use the distributive property to expand: $4(x + 3)$

Answer: _____

2. A rectangular garden bed is 3 m deep. Its width is split into two sections: x m planted with tomatoes and 5 m planted with peppers.

- (a) Write the total area of the bed, in square meters, as a *product* (depth times total width).

Answer: _____

- (b) Distribute to write the same area as a *sum* of the two section areas.

Answer: _____

- (c) In one sentence, explain how the two-section garden shows that your product and your sum are equal.

3. Combine like terms: $7x + 4 + 2x - 9$

Answer: _____

4. Factor out the greatest common factor: $6x + 15$

Answer: _____

5. Expand: $-2(4x - 7)$

Answer: _____

6. The three sides of a triangle measure x cm, $(x + 4)$ cm, and $(2x - 1)$ cm. Write an expression for the perimeter of the triangle, then combine like terms to simplify it.

Answer: _____ cm

7. 4 friends visit a deli. Each friend buys a sandwich that costs s dollars and a juice that costs 2 dollars.

(a) Write the total cost, in dollars, as a *product* (number of friends times cost per friend).

Answer: _____

(b) Distribute to write the total cost as a *sum*.

Answer: _____

8. Factor out the greatest common factor: $9x + 24$

Answer: _____

9. Simplify completely: $3(2x + 5) - 4x + 1$

Answer: _____

10. Challenge. Simplify completely: $5(2x + 3) - 2(x - 4)$

Answer: _____